

# Robot Sensors Reference

## **LiDAR**

LiDAR uses laser light to measure distances. It is useful for mapping, obstacle detection, and navigation.

## **IMU**

An IMU measures quantities such as linear acceleration and angular velocity. These measurements help estimate robot motion and orientation.

## **Wheel Encoders**

Wheel encoders measure wheel rotation and can be used to estimate wheel displacement, speed, and odometry.

## **Cameras**

Cameras provide image data for computer vision tasks such as object detection, tracking, inspection, and localization.

## **Ultrasonic Sensors**

Ultrasonic sensors estimate distance by transmitting sound waves and measuring the returning echo.